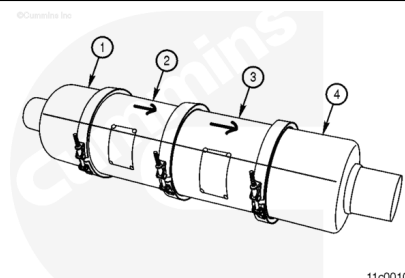


Trainings ▾

General Information

⚠ CAUTION ⚠

The catalyst elements contained in the aftertreatment system are made of brittle material. Do not drop or strike the side of the aftertreatment system, as damage to the catalyst element can result.



Due to the number of exhaust aftertreatment applications, this procedure is generic. **Not** all illustrations within the procedure will represent all applications.

The exhaust aftertreatment system is composed for four sections. These sections are:

1. Inlet
2. Aftertreatment Diesel Oxidation Catalyst (DOC)
3. Aftertreatment Diesel Particulate Filter (DPF)
4. Outlet

Note : In some applications, the catalyst can be integrated into the inlet of the exhaust aftertreatment.

Note : On some vertically oriented aftertreatment systems, the differential pressure sensor mounting bracket may be welded to the outlet section of the aftertreatment system. The outlet section of this type of aftertreatment system must be replaced if the differential pressure sensor mounting bracket is damaged.

Preparatory Steps

⚠ WARNING ⚠

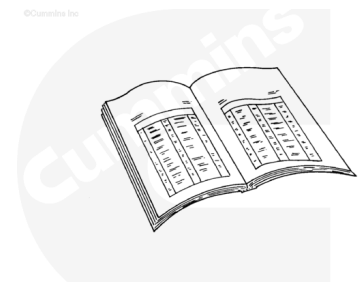
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,

remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

During regeneration, exhaust gas temperature can reach 800 °C [1500°F], and exhaust system surface temperature can exceed 700 °C [1300°F], which is hot enough to ignite or melt common materials, and to burn people. The exhaust and exhaust components can remain hot after the vehicle has stopped moving. To avoid the risk of fire, property damage, burns, or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they are likely to come in contact with hot exhaust or exhaust components.

- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Disconnect the aftertreatment wiring harness from the aftertreatment diesel particulate filter differential pressure sensor. Refer to Procedure 019-443 in Section 19. (</qs3/pubsys2/xml/en/procedures/10/10-019-443.html>)
- Remove the aftertreatment diesel particulate filter differential pressure sensor tubes. Refer to Procedure 011-047 in Section 11. (</qs3/pubsys2/xml/en/procedures/101/101-011-047-tr.html>)
- Remove the aftertreatment diesel particulate filter differential pressure sensor. Refer to Procedure 019-376 in Section 19. (</qs3/pubsys2/xml/en/procedures/16/16-019-376.html>) This procedure can be located in the Troubleshooting and Repair CM871 and CM876 Electronic Control System ISX and ISM Engines, Bulletin 4021560.
- Mark the orientation and location of the bracket before it is removed, to aid installation if there is **not** a locating tab present.



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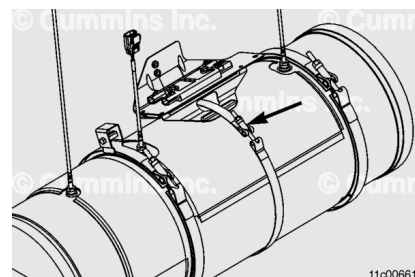
Remove

Clean and Inspect for Reuse

Install

⚠ CAUTION ⚠

The aftertreatment system must be installed so the aftertreatment DPF differential pressure sensor tubes slope downward to drain condensation away from the differential pressure sensor.



Note : A cylindrical locating tab may be welded to the outside of the catalyst canister to position the differential pressure sensor mounting bracket.

Note : Make sure the aftertreatment system is oriented so the aftertreatment DPF differential pressure sensor is installed in the same orientation noted during disassembly.

Install the differential pressure sensor mounting bracket and strap.

Be sure to align the cylindrical locating tab, if present, with the mounting bracket, as noted during removal.

If there is **not** a locating tab, use the reference mark that was made during removal.

Apply a coating of anti-seize compound to the threads of the aftertreatment DPF differential pressure sensor mounting bracket strap.

Install the nut on the aftertreatment diesel particulate filter differential pressure sensor mounting bracket strap.

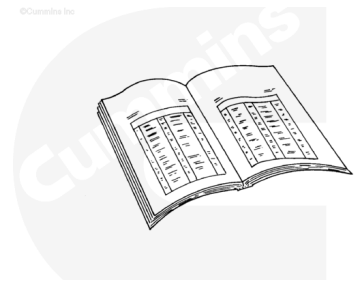
Tighten the differential pressure sensor mounting bracket strap.

Torque Value: 7 n·m [62 in-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Install the aftertreatment diesel particulate filter differential pressure sensor. Refer to Procedure 019-376 in Section 19. (</qs3/pubsys2/xml/en/procedures/16/16-019-376.html>) This procedure can be located in the Troubleshooting and Repair CM871 and CM876 Electronic Control System ISX and ISM Engines, Bulletin 4021560.
- Install the aftertreatment diesel particulate filter differential pressure sensor tubes. Refer to Procedure 011-047 in Section 11. (</qs3/pubsys2/xml/en/procedures/101/101-011-047-tr.html>)
- Connect the aftertreatment wiring harness from the aftertreatment diesel particulate filter differential pressure sensor. Refer to Procedure 019-443 in Section 19. (</qs3/pubsys2/xml/en/procedures/10/10-019-443.html>)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for proper operation.